

## Lecture 14: Double Integrals over General Regions

### Background & Motivation

Last time we integrated over rectangles. Most regions of interest—disks, triangles, regions between curves—aren't rectangular. Today we handle general regions using Type I and Type II setups, and introduce polar coordinates for circular symmetry. The Gaussian integral, which cannot be evaluated by single-variable methods, yields beautifully to the polar approach.

### 1. Warm-Up: Double Integral over a Rectangle

Evaluate  $\int_0^2 \int_1^3 (3xy^2) dy dx$ .

1. Inner integral ( $y$ ):  $\int_1^3 3xy^2 dy = 3x \left[ \frac{y^3}{3} \right]_1^3 = x(27 - 1) = 26x$
2. Outer integral ( $x$ ):  $\int_0^2 26x dx = [13x^2]_0^2 = 52$
3. Result: 52

### 2. Type I Regions (Vertical Slicing)

#### Definition 1: Type I Region

$$R = \{(x, y) : a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$$

$$\iint_R f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$$

- Integrate  $y$  first (inner), then  $x$  (outer)
- Think: vertical lines sweep left to right across the region

### 3. Type II Regions (Horizontal Slicing)

## Definition 2: Type II Region

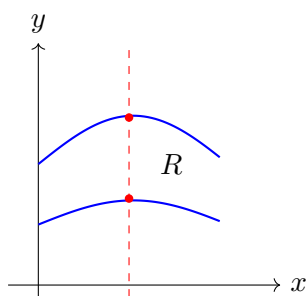
$$R = \{(x, y) : c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$$

$$\iint_R f(x, y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy$$

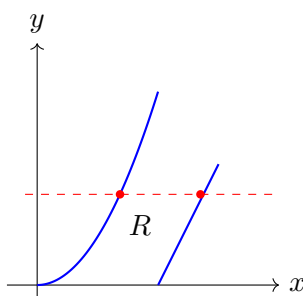
- Integrate  $x$  first (inner), then  $y$  (outer)
- Think: horizontal lines sweep bottom to top across the region

## How to Classify a Region

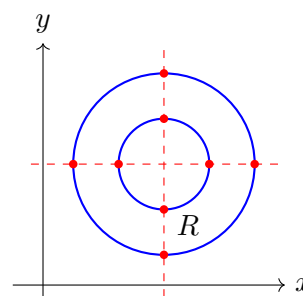
- Sketch the region and find all intersection points
- Type I – can every vertical line ( $x = c$ ) enter through one curve and exit through one curve?
- Type II – can every horizontal line ( $y = c$ ) enter through one curve and exit through one curve?
- Some regions work as both; pick whichever is simpler
- If neither works cleanly, **split** the region



Type I Region



Type II Region



Neither Type

A Type I region is bounded above and below by functions of  $x$  — every vertical line (red) hits the boundary exactly twice. Type II is analogous with horizontal lines. An annulus is *neither*: both horizontal and vertical slice lines can cross the boundary four times, so it must be split.

## 4. Examples

### Example 1: Type I: Region Between Two Parabolas

Evaluate  $\iint_R (x + 2y) dA$  where  $R$  is bounded by  $y = x^2$  and  $y = 2 - x^2$ .

1. **Sketch and intersections:**  $x^2 = 2 - x^2 \implies 2x^2 = 2 \implies x = \pm 1$ .

Lower curve:  $y = x^2$ . Upper curve:  $y = 2 - x^2$ .

2. **Set up (Type I):**

$$\int_{-1}^1 \int_{x^2}^{2-x^2} (x + 2y) dy dx$$

3. **Inner integral ( $y$ ):**  $[xy + y^2]_{x^2}^{2-x^2} = x(2 - x^2) + (2 - x^2)^2 - x \cdot x^2 - x^4$   
 $= 2x - x^3 + 4 - 4x^2 + x^4 - x^3 - x^4 = 2x - 2x^3 - 4x^2 + 4$

4. **Outer integral ( $x$ ):** Note  $2x - 2x^3$  is odd on  $[-1, 1]$  so integrates to 0.

$$\int_{-1}^1 (-4x^2 + 4) dx = 2 \int_0^1 (-4x^2 + 4) dx = 2 \left[ -\frac{4x^3}{3} + 4x \right]_0^1 = 2 \left( \frac{8}{3} \right)$$

5. **Result:**  $\boxed{\frac{16}{3}}$

### Example 2: Polar: Integral over a Disk

Evaluate  $\iint_D (x^2 + y^2) dA$  where  $D$  is the disk  $x^2 + y^2 \leq 4$ .

1. **Convert to polar:**  $x^2 + y^2 = r^2$ ,  $dA = r dr d\theta$ .

Disk:  $0 \leq r \leq 2$ ,  $0 \leq \theta \leq 2\pi$ .

2. **Set up:**  $\int_0^{2\pi} \int_0^2 r^2 \cdot r dr d\theta = \int_0^{2\pi} d\theta \cdot \int_0^2 r^3 dr$

3. **Evaluate:**  $= 2\pi \cdot \left[\frac{r^4}{4}\right]_0^2 = 2\pi \cdot 4$

4. **Result:**  $\boxed{8\pi}$

## 5. Polar Coordinates for Double Integrals

### Double Integrals in Polar

$$\iint_R f(x, y) dA = \int_\alpha^\beta \int_{r_1(\theta)}^{r_2(\theta)} f(r \cos \theta, r \sin \theta) \cdot r dr d\theta$$

– The extra factor of  $r$  comes from the area element:  $dA = r dr d\theta$

– Use polar when the region or integrand involves  $x^2 + y^2$ , circles, or sectors

### Example 3: The Gaussian Integral ( $I^2$ Trick)

Evaluate  $I = \int_{-\infty}^{\infty} e^{-x^2} dx$ .

1. **Square the integral:**  $I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2} dx\right) \left(\int_{-\infty}^{\infty} e^{-y^2} dy\right) = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dA$

2. **Convert to polar:**  $x^2 + y^2 = r^2$ , region is all of  $\mathbb{R}^2$ .

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} \cdot r dr d\theta$$

3. **Inner integral:** Let  $u = -r^2$ ,  $du = -2r dr$ :  $\int_0^{\infty} r e^{-r^2} dr = \left[-\frac{1}{2}e^{-r^2}\right]_0^{\infty} = 0 - \left(-\frac{1}{2}\right) = \frac{1}{2}$

4. **Outer integral:**  $I^2 = 2\pi \cdot \frac{1}{2} = \pi$

5. **Result:**  $I = \boxed{\sqrt{\pi}}$

*This integral cannot be evaluated by single-variable techniques — the  $I^2$  trick with polar coordinates is essential.*

## 6. Complex Regions (Splitting)

When a region is neither purely Type I nor Type II – **split** it:

$$\iint_R f \, dA = \iint_{R_1} f \, dA + \iint_{R_2} f \, dA$$

You can even use Cartesian for one piece and polar for another.

#### Example 4: Split Region: Cartesian + Polar

Evaluate  $\iint_R (1 + x^2 + y^2) \, dA$  where  $R$  is the region inside the square  $[-1, 1] \times [-1, 1]$  but outside the unit disk  $x^2 + y^2 < 1$ , combined with the upper half-disk  $x^2 + y^2 \leq 1, y \geq 0$ .

1. **Split:** Let  $R_1$  = upper half-disk ( $x^2 + y^2 \leq 1, y \geq 0$ ) and  $R_2$  = square minus full disk (corners).
2.  $R_1$  **in polar:**  $0 \leq r \leq 1, 0 \leq \theta \leq \pi$ .

$$\iint_{R_1} (1 + r^2) \cdot r \, dr \, d\theta = \int_0^\pi \int_0^1 (r + r^3) \, dr \, d\theta = \pi \left[ \frac{r^2}{2} + \frac{r^4}{4} \right]_0^1 = \frac{3\pi}{4}$$

3.  $R_2$ : The corners of the square outside the disk can be set up as a Cartesian integral:

$$\iint_{\text{square}} (1 + x^2 + y^2) \, dA - \iint_{\text{disk}} (1 + x^2 + y^2) \, dA$$

Square integral:  $\int_{-1}^1 \int_{-1}^1 (1 + x^2 + y^2) \, dy \, dx = \frac{20}{3}$ .

Full disk integral (polar):  $\int_0^{2\pi} \int_0^1 (1 + r^2)r \, dr \, d\theta = 2\pi \cdot \frac{3}{4} = \frac{3\pi}{2}$ .

Corners:  $\frac{20}{3} - \frac{3\pi}{2}$ .

4. **Combine:** Total =  $\frac{3\pi}{4} + \frac{20}{3} - \frac{3\pi}{2} = \frac{20}{3} - \frac{3\pi}{4}$

5. **Result:**  $\frac{20}{3} - \frac{3\pi}{4}$

## Practice Problems

**Problem 1.** Evaluate  $\int_0^1 \int_0^x xy \, dy \, dx$ .

Answer:  $1/8$

**Problem 2.** Evaluate  $\iint_R xy \, dA$  where  $R$  is bounded by  $y = 0$ ,  $y = \sqrt{x}$ , and  $x = 4$ . (Type I)

Answer:  $32/3$

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**Problem 3.** Evaluate  $\iint_D e^{-(x^2+y^2)} dA$  where  $D$  is the annulus  $1 \leq x^2 + y^2 \leq 4$ . (Use polar.)

Answer:  $\pi(e^{-1} - e^{-4})$

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**Problem 4.** Reverse the order of integration and evaluate:  $\int_0^1 \int_0^{\sqrt{y}} f(x, y) dx dy$  where  $f(x, y) = e^{y/x}$ .

Answer:  $\int_0^1 \int_{x^2}^1 e^{y/x} dy dx$

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**Problem 5.** Evaluate  $\iint_R \sqrt{x^2 + y^2} dA$  where  $R$  is the region in the first quadrant between  $x^2 + y^2 = 1$  and  $x^2 + y^2 = 9$ .

Answer:  $\frac{\pi}{2} \cdot \frac{26}{3} = \frac{13\pi}{3}$